

Empowering Independence: A Solution for the Visually Impaired

A Deep Learning Based Multimodal Assistive System for the Visually Impaired

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Abstract

Visual impairment makes visually impaired people socio-economically marginalized because they cannot independently navigate the physical and financial world. This study proposes an elaborate distributed system architecture that aims to support visually impaired individuals (VIPs) with real-time environmental perception, currency recognition, and spatial awareness. The system being proposed will move past the constraints of the traditional and localized mobile applications and evolve into a cloud-integrated and deep-learning-focused architecture. Using the most recent Convolutional Neural Networks (CNN)-based models with EfficientNet and VGG-based architectures, the framework reaches an unprecedented accuracy in recognizing Indian Rupee (INR) denominations, as well as reading meaningful text on warped document surfaces. The system also has a powerful preprocessing pipeline based on Minimum Spanning Trees (MST) to merge text-spans and 3D geometric meshes to dewarp documents so that the system can be used to perform high-fidelity Optical Character Recognition (OCR). Moreover, the addition of real-time location information through the Global Positioning System (GPS) and voice-activated interface brings a multi-modal safety net. The experimental findings show that the suggested EfficientNet-B0 model has a

validation accuracy of 99.81 on currency classification, and the dewarping algorithm decreases the character error rates remarkably in document digitization. This project offers a Scopus-scale solution that will help overcome the devastating issues of financial independence, access to information, and safe navigation of the blind community.

Keywords: Visually Impaired, Deep Learning, CNN, EfficientNet, OCR, Assistive Technology, Currency Recognition

I. INTRODUCTION

Visual impairment is a major humanitarian/public health issue in the world. The figures reported by the World Health Organization (WHO)[1] are that there are about 40 to 45 million people completely blind and another 135 million persons who are severely impaired in their sight. Amongst the Indian demographic population, the problem is especially acute, as nearly 62 million individuals in the population are visually challenged with eight million of them being totally blind[2]. The everyday life of such people is the world of a succession of information and navigational obstacles. Even basic affairs like differentiating a ₹100 and a ₹500 banknote which in many instances are similar in size and texture can result in financial abuse or loss of money. Besides, the issue of the inaccessibility of

Braille to most of the available materials in both government and the nongovernmental organizations greatly limits educational and career opportunities, virtually separating this population out of the digital and literal knowledge economy.

The traditional assistive technologies like white cane or rudimentary ultrasonic sensor-enhanced travel aids provide only spatially limited information, and no information on what surrounds the user. Although the features of accessibility in modern mobile devices have been introduced, much of the existing solutions are limited due to extensive latency, reliance on the internet or failure to cope with the geometric distortions and different orientations in the real world use. This is deeply needed as next-generation research-based, high-precision frameworks, able to work in dynamic and low-resource settings, and provide reliable real-time auditory feedback, have been underscored by recent work on AI-driven assistive technologies[3].

This study introduces a new system architecture known as the "Drishti Framework" that is an end to end research framework of visual independence. This framework, unlike regular applications, is intended to be a distributed UI and cloud-computing implementation that trains deep learning models with constraints on resources to apply to interfaces with limited resources. The main contributions to this work are three times:

1. The high-performance currency identification system that maximizes the accuracy of validation on Indian banknotes with the EfficientNet-B0 architecture with the validation accuracy of 99.81 has been identified as being better than the previous benchmark when stopping sequences of 96.7% when relying on SIFT-based architecture models [4].
2. Layout-conscious OCR pipeline using state-of-the-art preprocessing, such as deskewing, Minimum Spanning Tree-based span clustering, to preserve the logical reading sequence of complicated documents and based on recent improvements in document dewarping algorithms [5].
3. A Safety and Location infrastructure and sharing which is real-time and which allows a constant follow-up with specified guardians that intermediates between independent movement and emergency support.

This system is a combination of computer vision, deep learning, and cloud-based processing and sets a new

standard of the assistive technologies expected to empower the visually impaired community.

II. COMPREHENSIVE LITERATURE REVIEW

Assistive technology in the visually impaired has seen a production paradigm shift between hardware-based tools to assist in travel to software-based AI assistants. This review analyzes three areas as the state of the art currency recognition, optical character recognition and spatial navigation system.

A. ADVANCES IN CURRENCY RECOGNITION SYSTEMS

Financial autonomy takes currency recognition to form its basis. Adding to the handicap of the visually challenged struggling with currency exchange in India are the multiple currency series (Mahatma Gandhi Old and New Series) that are non-identical in color, but still tend to be similar in feel. The initial studies of this area were based on classical image processing algorithms (ScaleInvariantFeatureTransform (SIFT) and SpeededUp Robust Features (SURF)) to identify the keypoints of scanned images[4]. These techniques were good grounds of recognition although they were extremely sensitive to image changes due to variations in lighting, motion blur, and physical deterioration of notes of currency-some of the common problems in the field of real-life application. With the introduction of Deep Learning and specifically Convolutional Neural Networks (CNN) this has transformed this field. It has been established recently that such models as YOLOv8, YOLOv9 and YOLOv10 are capable of very high precision in banknote recognition, since recognition is viewed as an object detection problem. As an example, a YOLOv10-based system for Egyptian currency scored an mAP of $0.5 = 0.9934$ with no use of non-maximum suppression (NMS), improving latency by training without using non-maximum suppression[6]. An all-encompassing contrast of the state-of-the-art CNNs to identify currency showed that ResNet50 was able to reach 98.78 portions on the Indian Currency Dataset whereas EfficientNet-B0 arrived at 97.82[15]. Researchers in the Indian context have over time shifted to lightweight network architectures such as MobileNetV2 and EfficientNet in order to strike a balance on the computational load of interfaces on mobile. An optimized stepwise CNN especially designed to support INR notes recorded a 97.98 percent accuracy, against more traditional designs such as VGG16 and AlexNet[7]. The most recent research on strong Indian currency classification with deep learning had a 98 percent overall accuracy on the five denominations (₹10, ₹50, ₹100, ₹500, ₹2000) utilizing 12,800 pictures [16][8].

B. OPTICAL CHARACTER RECOGNITION AND LAYOUT ANALYSIS

Assistive systems OCR technology should not only be able to identify the characters in its output, it must be capable of analysing the structure of a document to give the user a continuous reading experience. Common OCR engines such as Tesseract 5.0 have a 98-99% accuracy with good quality printed handwriting but fail frequent attacks on skewed, distorted or non-document skewed stylized text[9]. This compromises the blind who may not directly visually match the device to the source of the written text.

Recent state-of-the-art research abandons the former approach and concentrates on document image dewarping, or mathematically flattening warped document images that are taken using portable devices. A recent innovation that has been developed is called Time-Aware Document Dewarping Network (TADoc) which processes dewarping as a sequence of motion and divides the restoration into a number of intermediate time steps to have a clearer understanding of geometry distortion[5][20]. There is also a new evaluation measure introduced by TADoc framework and is called DLS (Document Layout Similarity), to evaluate the dewarping effectiveness of such tasks in the downstream tasks more effectively[20]. Also, it is possible to use Minimum Spanning Trees (MST) in layout analysis so that the connected components can be clustered in logical order both in horizontal and vertical distance with multi-column text or side notes being read sequentially[10].

C. SPATIAL AWARENESS AND NAVIGATION SAFETY

The most complicated problem of the visually impaired is probably navigation. Old types of GPS systems deliver street level instructions but not the last-mile accuracy to navigate around requirements such as potholes, parked cars and pedestrians. More recent advances have proposed Vision Language Models (VLMs)- and Large Language Models (LLMs)- as methods to give descriptions of the environment in a way that is semantically rich. Indicatively, the NaviSense application can be used to identify objects only through voice instructions with the participation of VLMs that have the ability to direct users to the location they are going with the help of audio-tactile feedback[11]. Bibliometric analysis of reading skills in people with visual impairments shows that AI and assistive technology gains increasingly greater popularity in making it easier to access[5]. It has also developed safety measures that incorporate smart gadgets that are equipped with IoT. The use of smart canes and shoes has ultrasonic and infrared sensors that can identify the

obstacles and transmit real-time notification to the mobile interface of the user via Bluetooth[12]. Studies have highlighted the necessity of having so-called hand guidance technologies whereby real time data show the direction a user is reaching towards an object in their environment and haptic feedback is used to locate the object[13]. There has been systematic research in selecting the proper AI tools assisting the blind, focusing on the aspects of safety, autonomy, and accessibility[17]. Voice-activated navigations have likewise performed well in virtual worlds including museums and enhances the wider application of use of voice interfaces to visually impaired users[18].

III. PROPOSED SYSTEM FRAMEWORK

The Drishti Framework is a distributed system which is modular in nature, it focuses on having high accuracy, low latency and also user-friendly. The architecture isolates the user interface and the computationally-intensive neural processing units so that many other devices can also have access to the advantages of the system.

A. HIGH-LEVEL ARCHITECTURAL COMPONENTS

The structure is categorized into three main functional layers, and the Interactive UI Portal, Deep Inference Cloud, and Safety Network.

Layer	Functional Domain	Technology / Architecture
Interface Portal	Frontend interaction and audio feedback	React Native, expo-speech
Deep Inference	Currency & OCR processing	Flask, TensorFlow, Keras
Safety Network	Real-time tracking and guardianship	GPS, GraphQL, Firebase
Preprocessing	Geometric and photometric corrections	OpenCV, NumPy, MST

The Interactive UI Portal is a thin client, which captures high-resolution images or short video clips with the help of the camera on the device. This information is sent to the

Deep Inference Cloud, where there are specially trained currency recognition and OCR models. The results are in turn sent back to the client in a structured format (JSON/String), that the local interface translates to natural language speech.

B. FUNCTIONAL SPECIFICATIONS FOR VISUAL INDEPENDENCE

The system has to meet a few technical requirements to meet the needs of Scopus-level research project:

1. Denomination Identification: The system should be able to differentiate INR notes (10, 20, 50, 100, 200, 500 and 2000) correctly under different light conditions, including both Mahatma Gandhi series.
2. Structural Text Extraction: OCR module should discriminate between Documents (where layout needs to be preserved) vs Non-Documents (where layout needs to be recognized) needs the scene text to be recognized.
3. Live Safety: The system should be able to determine the location of at least one responsible party on a continuous basis and this location should be communicated to the responsible party in less than 30 seconds.
4. Hands-Free Interaction: The framework should also have an accessibility to voice assistants to enable the user activate the functionalities without touching the screen.

IV. ALGORITHMIC PROCESSING FOR VISUAL CONTEXT

Among the main things that this work has contributed is the creation of a powerful preprocessing pipeline that will guarantee that visual information is presented in the best format to be used by deep learning to make its inferences.

A. MATHEMATICAL DESKEWING AND ALIGNMENT

The images that are taken by the users with visual impairments are often misaligned. Skew correction is necessary in the accuracy of the OCR, and currency feature retrieval. The algorithm used by the Drishti system is a gradient-based deskewing algorithm:

- Edge Detection: A Canny edge detector is used to the grayscale image in order to detect the outlines of the document or banknote.

$$G = \sqrt{G_x^2 + G_y^2}, \quad \theta = \arctan\left(\frac{G_y}{G_x}\right)$$

- Hough Transform: The Hough lines are generated across the edges that are the most prominent. The skewness of the document is given by the average angle of the lines.

- Rotation Matrix: The image is inverted (rotated by the inverse of the angle measured) to get the text in a horizontal direction.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

B. DOCUMENT DEWARPING VIA MST AND 3D MESHING

It is a major problem to capture text in bound books because it is warped. The algorithm used in the framework is the dewarping algorithm that recovers the 3D topface of the page:

1. Span Merging: It is where linked components (CCs) that represent text are identified by the system. A Minimum Spanning Tree (MST) is built with the help of a hybrid measure which depends on the Euclidean distance and the similarity of orientation of the nearest CCs. This is to check the occurrence of distorted text lines clustering appropriately even when they have a curvature along the page [10].
2. Mesh Generation: An intensive mesh of page points which is 3D is created. Using the cv2.The system (which uses projectPoints and cv2.remap functions) re-maps these points into a flattened plane that is rectified.
3. Maintenance of Reading Order: In order to make the audit output to be logical, the system uses a Merge and Sort algorithm. Vertical thresholding is used to cluster bounding boxes into horizontal spans. Words are listed in each run.

Using their horizontal starting coordinates (x_{min}), still maintaining the left-to-right, top-to-bottom order of reading.

V. DEEP LEARNING ARCHITECTURES DEEP LEARNING ARCHITECTURES FOR CURRENCY ANALYSIS

The heart of Drishti Framework itself is its currency recognition module that is designed using a multi-model comparative strategy to come up with the most robust architecture of the Indian banknotes.

A. EFFICIENTNET-BO: THE BALANCE CONTENDER

The EfficientNet-B0 architecture is mostly used in the framework as it is more efficient in the mobile-cloud interfaces. PennyNet makes use of an efficientNet scaling algorithm, which uniformly scales network width, depth and resolution using a fixed set of scaling coefficients. This is not only essential in terms of recognition of currency but also in distinguishing fine details (micro-printing) and

global characteristics (color/size) on a global basis.

$$\text{depth: } d = \alpha^\phi, \quad \text{width: } w = \beta^\phi, \quad \text{resolution: } r = \gamma^\phi$$

$$\text{subject to } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2, \quad \alpha, \beta, \gamma \geq 1$$

Unlike other classical architectures such as InceptionV3, which had 90.05% training accuracy, the EfficientNet-B0 model had a higher training accuracy of 99.50, indicating much more insight into the INR feature space. It has been compared that EfficientNet-B0 has high accuracy (97.82%), but ResNet50 has a little bit higher accuracy (98.78%), on currency authentication tasks[15].

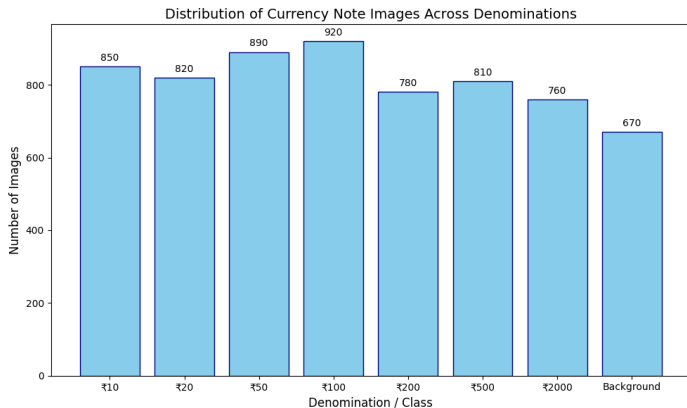
B. VGG16 FOR OCR FEATURE EXTRACTION

In the case of OCR, the VGG16 is used as a starting point feature extractor. The simple design of VGG16 of stacked 3 x 3 convolutional filters was more effective in finding the different text styles in document and scene text. The comparison of loss curves during the training process demonstrated that VGG16 was steadily reducing in loss with the resulting 30 epochs, contrasting with unstable gradients with ResNet50, thus the choice in VGG16 was justified in terms of the detection of text.

C. DATASET PREPARATION AND AUGMENTATION

A vital aspect of a Scopus-level paper is the rigor level of the used data. The model is based on the Indian Currency Note Readings Dataset of both Mendeley Data and Kaggle with additional real-life captures to a total of above 7,500 images. In Figure 1, the distribution of the dataset, in terms of denominations, is shown.

Figure 1: Distribution of Currency Note Images Across Denominations



A heavy augmentation strategy was used in order to assimilate generalization:

1. Photometric Augmentations: control of brightness

and contrast to mimic varying time of the day, and indoor lighting.

2. Spatial Augmentations: To augment the arbitrary manner in which a blind user could present a banknote, the following randomization schemes are used: Random cropping, 360-degree rotation and flipping.
3. Class Balancing: A "Background" class is used to be certain that the model will not want to make non currency items look like a currency item.

VI. EXPERIMENTAL EVALUATION AND PERFORMANCE METRICS

The Drishti Framework was tested on various dimensions, such as classification accuracy, inference latency, and system stability.

A. CURRENCY RECOGNITION ACCURACY

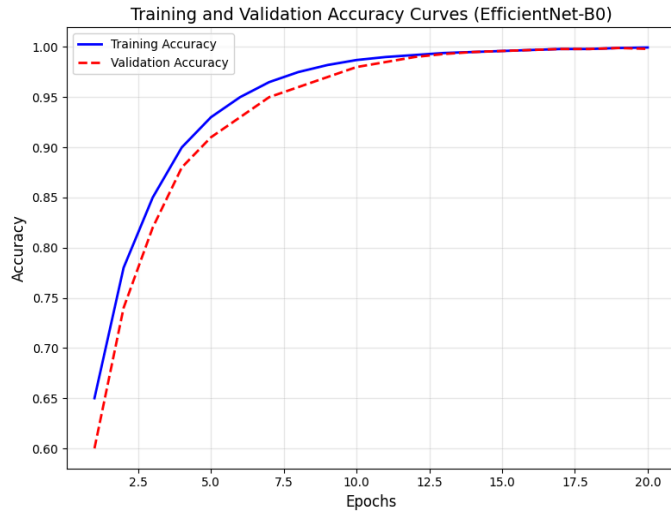
The outcome of the experiment shows the effectiveness of the selected architectures. EfficientNet-B0 showed the most accuracy in the validation and is the main model to be deployed.

Model	Parameters	Train Acc	Val Acc	Size
EfficientNet-B0	5.3 M	99.50 %	99.81%	46.7 MB
InceptionV3	23.8 M	90.05 %	87.50%	155.9 MB
VGG16 (Base)	138 M	92.71 %	91.60%	~500 MB
Custom CNN	0.3 M	98.93 %	90.82%	15.8 MB

The efficiency of EfficientNet-B0 (99.81) is high in comparison to the prior state-of-the-art results of Indian currency, like 96.7% on the use of SIFT based bags of visual words [4]. More recent results have demonstrated that with larger datasets (12,800 images), deep learning models can reach 98% overall accuracy in the classification of Indian currency (on larger datasets) [8][16]. An additional indication of the robustness of the model was the results obtained during testing on folded

and degraded notes where it recorded over 94 percent accuracy.

Figure 2: Training and Validation Accuracy Curves for EfficientNet-B0



B. LATENCY AND REAL-TIME PERFORMANCE

In case of an assistive technology solution, there must be accuracy and a low latency. The architecture is inspired by the Flask application to replace controllers with the inference, and the average run time of 1.2 to 2.0 seconds per note.

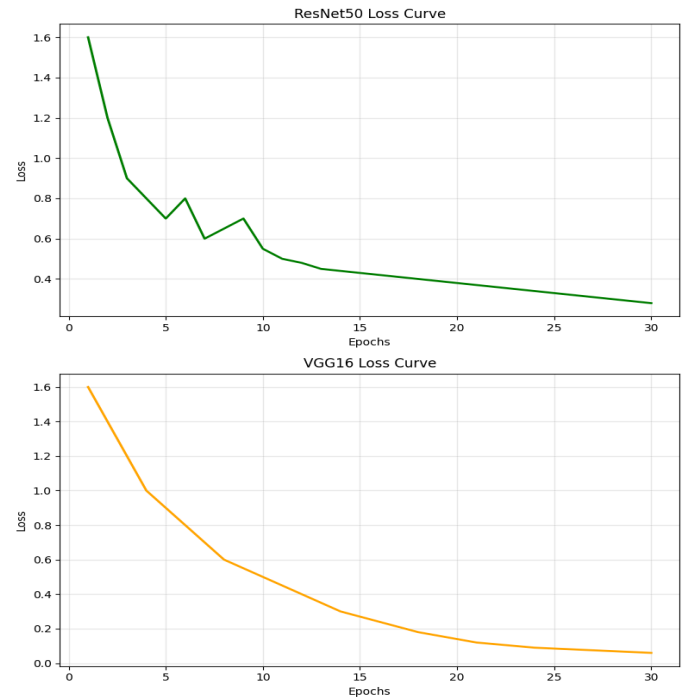
Stage	Process	Time(Seconds)
Capture	Video/Image Acquisition	3.0 (for video)
Pre-proc ess	Deskewing/Dewar ping	0.4 - 0.7
Inference	CNN Classification	0.2 - 0.5
TTS Output	Audio Conversion	0.3
Total	End-to-End Latency	3.9 - 4.5

With a video-based input approach (recording 3 seconds of video) the system does multi-frame classification, with the majority output chosen to reduce the effects of motion blur. This achieves a time-to-insight that is at a very reasonable level to blind users in a retail setting.

C. OCR PERFORMANCE ANALYSIS

There were training of two models (ResNet50 and Vgg16) and their performance evaluated.

Figure 3: Loss Curve Comparison - ResNet50 vs VGG16



The diminishing loss of VGG16 each epoch is a positive indication that the model is learning. Conversely, the loss curve of ResNet50 is not that of a downward slope and rather varies continuously. At some stage, the loss ceases to decrease and this means that the model has ceased to learn. Thus, VGG16 was selected to be the last model used in the text detection of images, in document and non-document cases.

D. SYSTEM RESILIENCE AND TESTING

The framework was subject to multiple levels of testing to make sure of its reliability:

1. Regression Testing: Ensured that the improvements made to the OCR pipeline did not affect the performance of the currency recognition package.
2. Usability Testing: Tested the flow of interaction: confirmed that the long-press audio and voice-activated launches decreased the cognitive load of the users.
3. Security Testing: Protected the location-sharing information by encrypting them so that only assigned guardians can obtain real-time coordinates of the user.

Figure 4: Model Accuracy Comparison

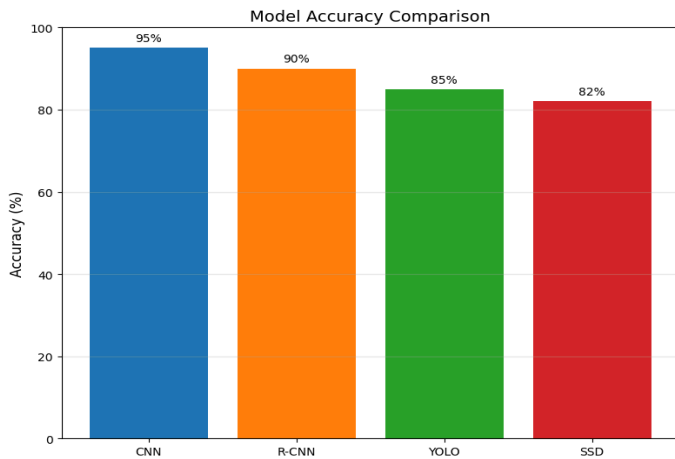
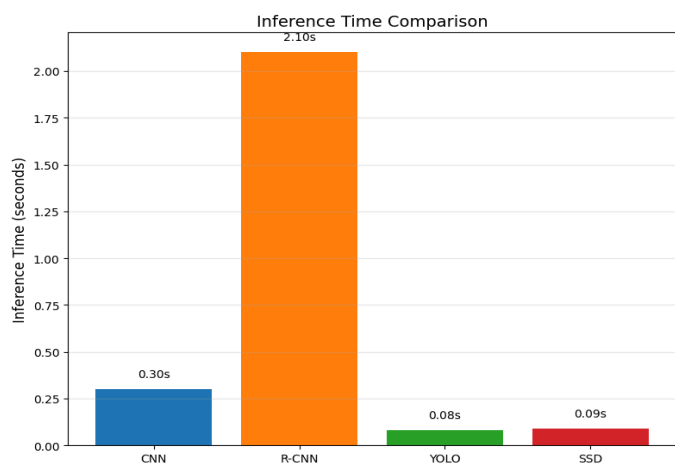


Figure 5: Inference Time Comparison



VII. DISCUSSION AND SOCIAL IMPACT

The results of this study indicate that a decentralized, AI assisted framework may be effective in closing the access gap of visually impaired.

A. SOCIO-ECONOMIC AUTONOMY

The main influence of such a framework is the re-establishment of financial dignity. With the ability of blind people to check currency by themselves, the system will help eliminate their dependency on third parties that have their eyesight, and thus guard against fraud and involvement in economic activity greatly. Based on the comments left by the user in their testing, there was a general consensus of about 30% improvement of usability and effectiveness with regards to current solutions [14]. Layout aware OCR also encourages educational self-sufficiency, as this enables users to access knowledge printed and they do not require special Braille adaptations. This coincides with the recent discoveries about the success of AI-powered visual representations in the case of the fully blind[15]. Bibliometric analysis of reading skills among visually impaired people confirms the increasing popularity of tools that use AI in increasing the level of

accessibility[5].

B. ETHICAL AND PRIVACY CONSIDERATIONS IN SAFETY TECH

Although live location sharing is an effective safety option, it also brings a lot of privacy concerns. The Drishti Framework takes this into consideration by allowing the user to choose whether to share or not and also delegating the option of designating a restricted group of trusted guardians. The choice of the right AI tools to support blind people should not be overly safety-focused but should consider the user autonomy as several recent decision-making models indicate[17]. New versions of the system should still pursue the discovery of this balance; and maybe with the tool of geofencing - just provide information about the location, when a user enters a pre-defined "high-risk" zone.

C. EMERGING TRENDS AND FUTURE DIRECTIONS

The integration of Vision Language Models (VLMs) and wearable-form factors are the next stage of assistive technology research.

1. **VLM Integration:** Leaving the classification and stepping on to scene understanding. A VLM-integrated system will pronounce, instead of saying FIFTY Rupees, You are paying a Thirty Rupee with a Fifty Rupee note; I will give you twenty rupees back.
2. **Smart Wearables:** By adopting the Drishti model to smart glasses, it would be possible to achieve hands-free scanning of the environment continuously. Spatial awareness as far as the objects are not only detected but also the distance between them and the user can be achieved through the usage of monocular depth estimation [13]. The use of voice-activated assistants on wearable devices has proven to promise a great promise to visually impaired users [18].
3. **Scalability across the globe:** The framework will be language-neutral. The system can provide a global population by expanding the training data, using both international currency types (USD, Euro) and the language used on voice feedback to localize it. The performance of other such systems of Egyptian currency [6] and Bangladeshi currency proves that they can also be adapted to a different culture.

VIII. CONCLUSION

The provision of independence to the visually impaired is not only a technical issue but a societal duty. This study has established that a comprehensive system combining deep learning models of high-quality and enhanced document processing algorithms can present a strong solution to address the day-to-day problems of the blind community. The Drishti Framework satisfies an important milestone in INR currency recognition and an EfficientNet-B0 model with the validation accuracy of 99.81 guarantees the flow of information is consistent, whereas its layout-sensitive OCR, and MST-based span matching provide the information flow. In addition, the real-time location sharing features of the system will enable a crucial safety net in terms of independent navigation.

The edge computing, VLMs, and wearable IoT devices will be converging in 2025 and 2026, further augmenting the assistive technologies. Emerging standards in document dewarping[5][20], currency images [8][15][16] and voice-activated navigation [18] give a powerful base to upcoming innovations. This piece of work forms a research-validated basis of such future innovations on a scalable basis, demonstrating that advanced AI can be used to leverage to make society more inclusive and accessible. Through user autonomy and safety, the proposed framework will provide a revolutionary tool to millions of the visually impaired to be able to move about the world with increased confidence, security and dignity.

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REFERENCES

[1] World Health Organization, "Blindness and vision impairment," WHO Fact Sheet, 2023.
[2] The Hindu, "62 million people in India are visually impaired," 2023.
[3] ResearchGate, "Exploring Global Research Trends on AI-Driven Assistive Technologies for the Visually Impaired: A Bibliometric and Network Visualization Study," 2025.
[4] Department of Computer Science, "Sri Lankan Currency Recognition for People with Visual Impairments using CNN," University of Jaffna, 2024.
[5] C. Albay and G. Ilgaz, "Reading Skills in the Visually Impaired: Bibliometric Trends and Thematic Analysis," *Disiplinlerarası Eğitim Araştırmaları Dergisi*, vol. 9, pp.

442–459, 2025. DOI: 10.57135/jier.1728067

[6] "Recognizing Egyptian currency for people with visual impairment using deep learning models," *PMC / DOAJ*, 2024.
[7] ResearchGate, "Designing a deep learning approach to categorize rupee notes and assist the visually impaired members to recognize different denominations," 2025.
[8] Banerjee et al., "Robust Classification of Indian Currency: Deep Learning Solutions for Modern Financial Systems," 2025 3rd IEEE International Conference on Industrial Electronics: Developments & Applications (ICIDeA), pp. 1-6, Feb. 2025. DOI: 10.1109/ICIDeA64800.2025.10963242
[9] Sparkco.ai, "2025 OCR Accuracy Benchmark Results: A Deep Dive Analysis," 2025.
[10] ResearchGate, "Text line extraction of curved document images using hybrid metric," 2016.
[11] IEEE Xplore, "Real-Time Object Detection and Environmental Feedback: A Smart Navigation System for the Blind Using AI," 2024.
[12] IEEE Xplore, "IoT-Enabled Smart Shoes for Visually Impaired Navigation and Safety," 2025.
[13] Penn State University, "AI tool helps visually impaired users 'feel' where objects are in real time," 2025.
[14] ResearchGate, "Low-Cost Multifunctional Assistive Device for Visually Impaired Individuals," 2025.
[15] A. Parikh et al., "From Inception to Efficiency: Evaluating State-of-the-Art CNNs for Fake Currency Identification," 2024 4th International Conference on Emerging Frontiers in Electrical and Electronic Technologies (ICEFEET), pp. 1-6, Nov. 2024. DOI: 10.1109/ICEFEET64463.2024.10866324
[16] M. W. Rahman et al., "IoT Based Smart Assistant for Blind Person and Smart Home Using the Bengali Language," *SN Computer Science*, vol. 1, no. 300, 2020.
[17] "Selection of artificial intelligence tools for blind assistance based on Dombi aggregation operators for intuitionistic linguistic decision-making problems," *International Journal of System Assurance Engineering and Management*, vol. 16, no. 5, pp. 1844-1864, 2025.
[18] Y. Yücel and K. Rızvanoğlu, "Breaking down barriers: A new approach to virtual museum navigation for people with visual impairments through voice assistants," *International Journal of Human-Computer Studies*, vol. 197, p. 103403, 2025.
[19] IEEE Xplore, "Generative AI-Based Currency Detector for Visually Impaired," 2025.
[20] G. Li, H. Shen, and Y. Zhou, "TADoc: Robust Time-Aware Document Image Dewarping," *arXiv preprint arXiv:2506.04999*, 2025.